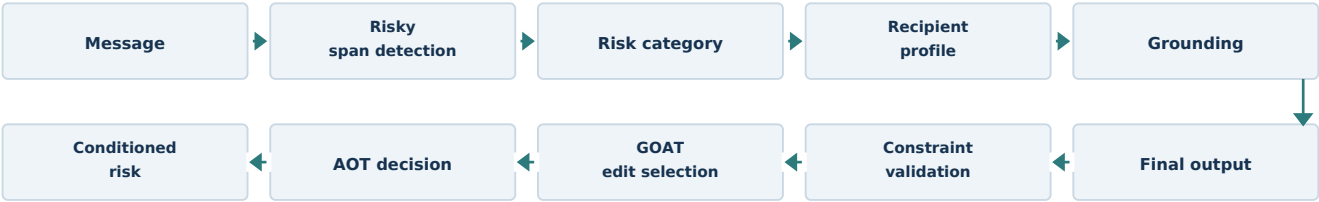


AOT-GOAT Explainability Analysis

Intrinsic, pipeline-level traceability of recipient-conditioned interventions

1. How Decisions Are Explained



AOT-GOAT is explainable at the pipeline level because the intervention decision can be traced through explicit intermediate variables: detected spans and categories are combined with recipient history and grounding evidence to form recipient-conditioned risk; AOT then selects PASS or INTERVENE; when intervention occurs, GOAT limits downstream editing and the result is checked against constraints. No formal SHAP/LIME-style post-hoc attribution is claimed.

2. PASS Example

Repository-documented implementation scenario - not a claimed evaluation row.
Message: "Please send it to everyone by EOD tomorrow, as discussed. The SLA needs sign-off before the meeting."
Sender evidence defines it as the Q3 budget report, everyone as Asha/Ravi/Karan, as discussed as Monday's budget discussion, the meeting as Friday's budget review, and SLA as service-level agreement. Recipient A is described as having extensive shared history containing these concepts, with a possible AOT PASS outcome.

Why PASS is explainable: the scenario provides recipient-specific shared history/evidence that grounds the potentially ambiguous references, so the documented gate outcome preserves the original message.

3. INTERVENE Example

Same message, different recipient context.
Recipient B is described as a newer contact with limited shared history; the documented possible outcome is AOT INTERVENE followed by GOAT selection of several high-risk spans. Recipient C is described as understanding some but not all context; the possible outcome is also INTERVENE with only necessary edits. This illustrates recipient-conditioned decision behaviour without claiming these are scored eval_set rows.

4. GOAT Minimal-Edit Example

Stage	Repository-supported interpretation
Original message	Full sender message enters the adaptation pipeline.
Detected spans	Potentially risky spans such as it, everyone, EOD tomorrow can be identified by the rule/lexicon layer.
Selection	GOAT is designed to select only the necessary risky spans rather than rewrite the complete message.
Adaptation + validation	Evidence-grounded replacement is attempted, then constraint validation determines whether the edit is accepted or fallback handling is used.

Important data boundary: raw GOAT candidate/edit objects, selected span ranking and numeric per-span scores were not retained in the accessible evaluation record, so no fabricated replacement string or ranking is presented.

AOT-GOAT Explainability Analysis

Evidence trace, examples and explainability finding

5. Explanation Table

Message span	Risk type	Recipient evidence	Risk / score	AOT	GOAT action
it	Coreference	Q3 budget report established in sender evidence	Not retained	PASS*	None
everyone	Scope	Asha, Ravi, Karan established in sender evidence	Not retained	PASS*	None
EOD tomorrow	Temporal / terminology	Timing context documented; recipient familiarity described	Not retained	INTERVENE*	Select high-risk span(s)*
SLA	Terminology	Service-level agreement definition documented	Not retained	Recipient-dependent*	Minimal edit if required*
as discussed	Implicit Context	Monday's budget discussion documented	Not retained	Recipient-dependent*	Minimal edit if required*

* Decision/action labels are taken from the repository-documented implementation scenario, not from a scored eval_set row. Raw per-span numeric scores and edit objects are not retained in the accessible record.

6. Key Finding

AOT-GOAT provides a traceable pipeline-level explanation for recipient-specific intervention. The implementation exposes a chain from concrete ambiguity spans and categories to recipient evidence, conditioned risk, the AOT gate, and downstream GOAT editing. Explainability is therefore intrinsic to the decision pipeline rather than dependent on a separate post-hoc attribution method.

Evidence trace in one view

Message state	Recipient evidence	Conditioned risk	AOT	GOAT	Validated output
"Send it..."	Shared entity / history	Lower when grounded	PASS	No edit	Original retained
Same message	Limited history / weaker grounding	Higher when evidence is insufficient	INTERVENE	Minimum edit set	Adapted or fallback

7. Limitations for Academic Interpretation

- The strongest concrete examples in the retained record are repository-documented scenarios; raw eval_set case IDs and per-case outputs are not available here.
- No formal SHAP/LIME-style attribution is reported; the explainability claim is pipeline-level/intrinsic.
- The deterministic baseline is English-focused and rule/lexicon driven, so explanations are strongest where detected ambiguity aligns with explicit patterns and context evidence.
- Sender-provided context is treated as evidence by the design; the system does not independently verify that evidence.
- Production deployment would require additional controls such as authentication, authorization, input validation, rate limiting and secret protection.